

Question 4

1. Strategy Design

Strategy: Mean-Reversion + Momentum

Concept: This strategy actively exploits short-term price extremes while trading around a baseline average. It aims to buy when the stock has statistically dropped too far (oversold) and sell when it statistically spikes too high (overbought).

Indicators Used (4):

1. **SMA20 (Simple Moving Average - 20 Period):** Defines the intermediate-term baseline price.
2. **Lower Bollinger Band (SMA20 - 2 StdDev):** Identifies strong downward price deviations.
3. **Upper Bollinger Band (SMA20 + 2 StdDev):** Identifies strong upward price deviations.
4. **RSI (Relative Strength Index - 14 Period):** Measures the momentum of the price moves.

```
# indicator 1, 2 & 3: Bollinger Bands (SMA20 + Upper/Lower Std Dev)
df['SMA20'] = df['Close'].rolling(window=20).mean()
df['STD20'] = df['Close'].rolling(window=20).std()
df['UpperBand'] = df['SMA20'] + (df['STD20'] * 2)
df['LowerBand'] = df['SMA20'] - (df['STD20'] * 2)

# indicator 4: RSI
df['RSI'] = calRSI(df['Close'], window=14)

# indicator 5: Volume SMA
df['VolAvg'] = df['Volume'].rolling(window=20).mean()
df['Signal'] = 0
```

Trading Rules:

- **BUY when:**
 - Close Price < Lower Bollinger Band
 - RSI < 40

(Logic: The stock has sharply deviated below its normal range and momentum is sufficiently oversold).
- **SELL when:**

- Close Price > Upper Bollinger Band

- **OR** RSI > 70

(Logic: The stock has sharply surged above its normal range or momentum is overbought, prompting an exit).

2. Trade Sheet

Below are 15 sample trades generated by the strategy during the test and train:

`TestHDFC.csv`

Entry Date	Entry Price	Exit Date	Exit Price	Return (%)
2021-04-12	647.42	2021-05-18	699.35	8.02
2021-07-20	686.42	2021-08-10	717.09	4.47
2021-09-22	729.49	2021-09-24	761.76	4.42
2021-12-20	678.09	2022-01-05	744.30	9.76
2022-02-24	675.12	2022-03-29	690.53	2.28
2022-09-26	686.46	2022-10-31	720.16	4.91
2023-01-10	754.62	2023-01-23	805.04	6.68
2023-03-15	741.91	2023-04-05	795.73	7.25
2023-05-05	782.21	2023-06-22	799.92	2.26
2023-10-25	728.33	2023-11-16	734.09	0.79
2024-01-17	748.28	2024-04-02	720.37	-3.73
2024-05-09	704.48	2024-05-30	747.32	6.08
2025-01-06	843.84	2025-02-04	848.97	0.61
2025-08-28	957.80	2025-09-18	976.90	1.99

Others can be found in the `Trades.pdf` or visit the folder `Q4-Trades` and see the corresponding CSVs to get the same.

3. Evaluation Metrics

Training Evaluations

Stock	Total Trades	Total Return (%)	Sharpe Ratio	Max Drawdown (%)	Win Rate (%)
INFY	30	57.20	0.29	-21.63	66.67
MARUTI	24	8.87	0.04	-32.31	54.17
HDFC	28	59.71	0.39	-15.74	67.86
SUNPHARMA	35	96.46	0.31	-30.51	68.57

Corresponding CSV files having Data and Trade records can be found in [Data-Split/Train/](#) and [Q4-Trades/](#) respectively.

Testing Evaluations

Stock	Total Trades	Total Return (%)	Sharpe Ratio	Max Drawdown (%)	Win Rate (%)
INFY	11	-36.72	-0.36	-34.63	45.45
MARUTI	11	23.53	0.61	-6.85	81.82
HDFC	14	55.82	1.12	-3.73	92.86
SUNPHARMA	13	31.03	0.57	-3.27	69.23

Corresponding CSV files having Data and Trade records can be found in [Data-Split/Test/](#) and [Q4-Trades/](#) respectively.

One can easily see that returns have been dropped in the testing than they were in the training, which suggest that we over-fitted to our assumptions and unseen data is causing issue in the future (Luckily MARUTI gave more returns in testing).
